Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit.

1. If the p-value is very small, which of the following is true?

   (a) Very strong evidence exists against the null hypothesis.
   (b) The alternative hypothesis has been proven.
   (c) Very strong evidence exists against the alternative hypothesis.
   (d) The null hypothesis has been proven.

2. Most experts recommend college students get at least 8 hours of sleep per night. A random sample is collected of 112 students at a college by asking how many hours of sleep each student got last night. The sample mean amount of sleep last night was 7.6 hours, with a sample standard deviation of 2.1 hours. The sample distribution of hours of sleep was slightly skewed right.
   Use the sample results to estimate the standard error of the sampling distribution of sample means.

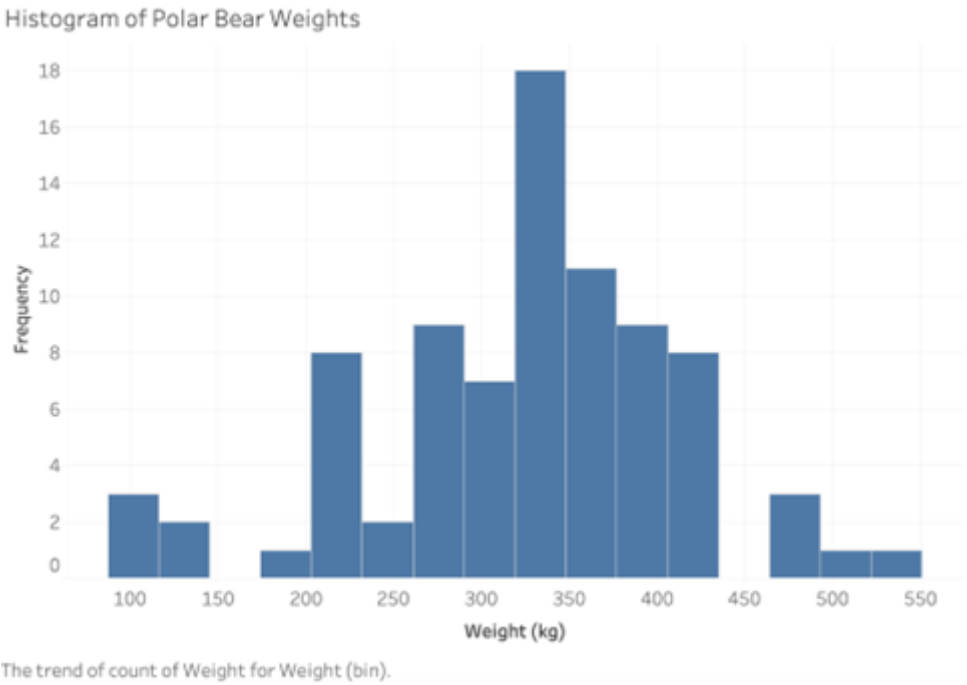
   (a) 0.718
   (b) 0.198
   (c) 2.1
   (d) 7.6

A hot dog manufacturer samples 10 hot dogs and calculates the average fat content to be 18 grams per hot dog and a standard deviation of 1.1 grams. We know from past studies that fat contents in hot dogs from this manufacturer follow an approximately normal distribution.

3. The value 18 is the _____.

   (a) sample size
   (b) statistic
   (c) standard deviation
   (d) parameter

Adult male polar bears are expected to weigh, on average, 370 kg, but there is concern by some scientists that because of recent warmer winters that polar bears might be struggling to gain weight. The US Geological Survey (USGS) conducted a study in the Southern Beaufort Sea on the weight of polar bears. Eighty-three adult male polar bears were captured between March and May of the years 1990 and 2006 and their weights were recorded. The sample mean weight was 324.6 kg, and the sample standard deviation was 88.3 kg.
A histogram of the 83 polar bear weights in the sample is shown below.



4. Are the validity conditions for theory-based inference met for these data?

   (a) No, since the sample distribution is slightly skewed left.
   (b) Unable to determine.
   (c) Yes, since at least 10 polar bears weighed more than 370 kg and at least 10 weighed less than 370 kg.
   (d) Yes, since the sample size is larger than 20 and the sample distribution is not strongly skewed.

5. What are the appropriate null and alternative hypotheses?

   (a) $H_0 : \overline{x} = 370$ versus $H_A : \overline{x} \neq 370$
   (b) $H_0 : \mu = 370$ versus $H_A : \mu < 370$
   (c) $H_0 : \overline{x} = 370$ versus $H_A : \overline{x} < 370$
   (d) $H_0 : \mu = 370$ versus $H_A : \mu \neq 370$

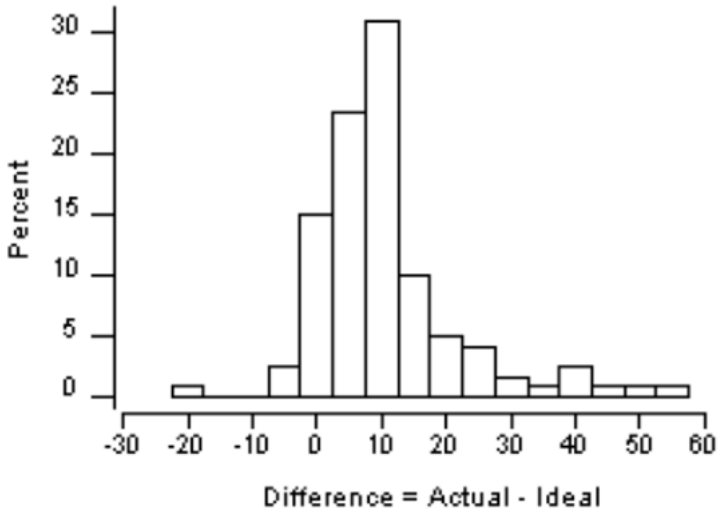
6. The sample mean of 324.6 kg lies _____(1)_____ standard errors _____(2)_____ the expected average polar bear weight of 370 kg.

   (a) 0.51, below
   (b) 0.51, above
   (c) 4.68, below
   (d) 4.68, above

7. A theory-based inference applet gives a p-value less than 0.00001. Do these results provide evidence that the true mean weight of polar bears is less than 370 kg?

   (a) No, since the sample results could have happened by chance.
   (b) Yes, since the p-value is small.
   (c) No, since the p-value is large.
   (d) Yes, since 324.6 kg is less than 370 kg.

The following histogram shows the distribution of the difference between the actual and "ideal" weights for 119 female students. Ideal weights are responses to the question "What is your ideal weight"? The difference = actual – ideal.



8. What is the approximate shape of the distribution?

   (a) Skewed left
   (b) Extremely skewed right
   (c) Unable to determine
   (d) Approximately normal with slight skew to the right

9. How does the mean compare to the median in this data set?

   (a) The median is larger than the mean.
   (b) The median and mean are approximately equal.
   (c) The mean is larger than the median.
   (d) The mean and the median cannot be compared without calculations.

10. If the observation at -20 pounds was removed from the data set, the mean _____.

    (a) would remain the same
    (b) may or may not change
    (c) would increase
    (d) would decrease

11. Which of the following distributions would have the smallest standard deviation?

    (a) Average number of miles put on a rental car per day across 25 customers
    (b) Number of miles put on a rental car in one day
    (c) Distributions of all of the the other three answers would have the same standard deviation.
    (d) Average number of miles put on a rental car per day across 5 customers